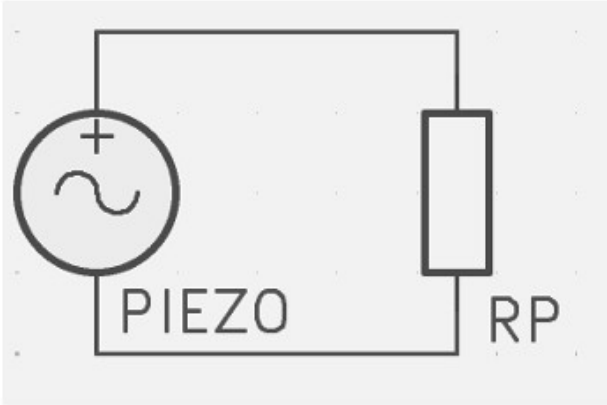


LSD\_UltraS circuit documentation & derivation

This documentation derivations used to make the LSD\_UltraS distance sensor.  
The main device is a piezo transducer, the physics of the transducer shall be defined before trying to boost it since depending on how it drives current will decide how the circuit needs to be setup.

The piezo

A piezo transducer is physically a crystal with electrodes, its electrical behaviour can be described using the Butterworth-Van Dyke model. At 40KHZ the piezo acts as a open voltage source.

|                                                                                                                                                                                                                                                                       |                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| PIEZO definition                                                                                                                                                                                                                                                      |                                                                                     |
| $v_{piezo} = open\ piezo\ putput$ $R_S = resistance\ real$ $X_S = reactance\ imaginary$ $Z_{piezo} = impedance$<br>$Z_{piezo} = R_S \pm jX_S$ $Z_{piezo} \approx R_S\ (at\ resonance)$<br>$i_P = \frac{V_P}{Z_{piezo} + R_P}$<br>$v_{RP} = \frac{V_P R_P}{Z_P + R_P}$ |  |

Transistor signal booster

To boost the signal to a readable digital value the component of choice would be a transistor, to set up the transistor the DC bias must be set for it to be in operating range. Once the DC bias is set the small signal AC model can be defined to solve for the required values of gain and saturation points.

| DC analysis                   | AC analysis                              |
|-------------------------------|------------------------------------------|
| Find operating point(Q point) | Finds gain, impedance, freq response     |
| Uses full circuit             | Small signal model with DC source as GND |
| Gives saturation points       | Give how signal gets amplified           |
| Yields: IC,VCE, VB,VE, VC     | Yields: A, Rin, Rout                     |

## Setting up (DC)

### TRANSISTOR STATE:

The transistor is Current driven where the Gain  $\beta$  is the ratio between the current passing through the base emitter to the collector emitter. For this circuit the input variables will be the base current and the output variables are the Collector current. Defining the gain relation to the transistor the DC levels of the transistor can be set using KCL, the AC part of the circuit is to be handled with separately for any linear gain amplification.

### TRANSISTOR SETUP

$$I_E = I_C + I_B$$

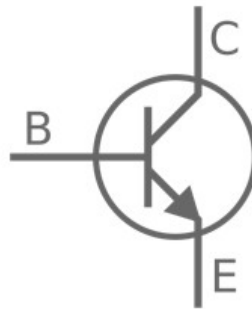
$$\frac{I_C}{I_B} = \beta$$

$$\frac{I_E}{I_C} = \left( \frac{I_C + I_B}{I_C} \right) = \left( 1 + \frac{1}{\beta} \right) = \left( \frac{1 + \beta}{\beta} \right)$$

$$\frac{I_E}{I_C} = \left( \frac{1 + \beta}{\beta} \right) = (1 + \beta) \frac{I_B}{I_C}$$

$$I_E = (1 + \beta) I_B$$

$$I_C = \beta I_B$$



### KCL(DC)

#### Loop 1 (find $I_C$ )

$$V_{CC} = I_B R_B + V_{BE} + I_E R_E$$

$$V_{CC} = I_B R_B + V_{BE} + (1 + \beta) I_B R_E$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E}$$

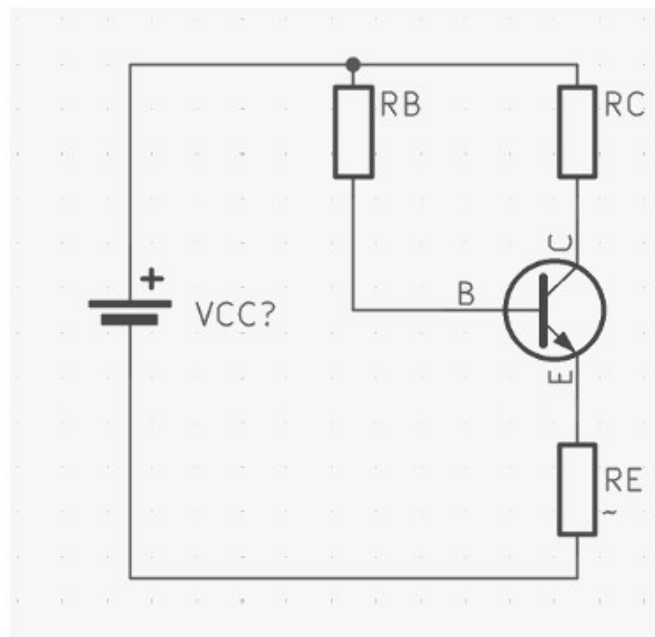
$$I_C = I_B \beta = \left( \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E} \right) \beta$$

#### Loop 2 (find $V_{CE}$ )

$$V_{CC} = I_C R_C + V_{CE} + V_E$$

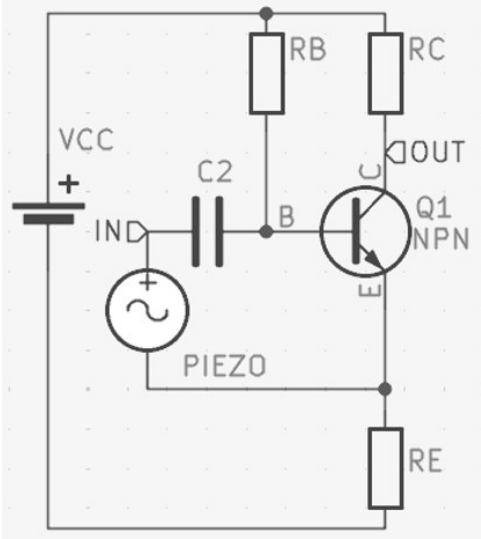
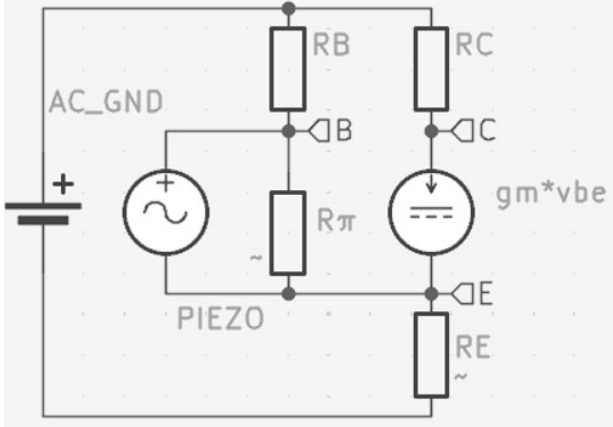
$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$V_{CE} = V_{CC} - I_B \beta R_C - (1 + \beta) I_B R_E$$



## Setting up AC

The AC side of the circuit is redefined to solve small signal gain around the DC operating point.

| CIRCUIT                                                                           | AC equivalent                                                                      |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |

## TRANSISTOR (Ebers-Moll model)

$$i_c = I_s e^{\frac{v_{BE}}{v_t}}$$

$$e^{\frac{v_{BE}}{v_t}} = \text{exponential factor}$$

$$I_s = \text{saturation current}$$

$$v_T = \text{thermal voltage} - (\text{diodes \& transistors})$$

**Key:** the transistor isn't a linear device, & the exponential behavior of the physics behind silicon is what causes the amplifying power and may be influenced by temperature.

## Small-Signal Model Setup (standard NPN)

$$g_m = \text{Transconductance } dI/dv$$

$$i_c = I_s e^{\frac{v_{BE}}{v_t}}$$

$$g_m = \frac{di_c}{dv_{BE}} = \frac{d\left(I_s e^{\frac{v_{BE}}{v_t}}\right)}{dv_{BE}} = \frac{I_s e^{\frac{v_{BE}}{v_T}}}{v_T} = \frac{i_c}{v_T}$$

**Key:**  $g_m$  expresses the output current per input voltage, where output is the current passing through the collector and input is the voltage across VBE. In simplified terms the higher the DC Bias Current is set the more sensitive the output is to small input signals.

## Interpreting the AC circuit

1. DC rails are treated as AC GND
2. Capacitor can be treated as a short (for solving gain)
3. Base emitter denoted as  $R_\pi$
4. Collector emitter denoted as current supply ( $gm v_{be}$ )

Using Ebers-Moll model the gain for small signal model can be used to find transconductance, this allows the equation to be solved algebraically since now the transistor is defined by 2 components which act as a current supply between CE and a resistor between BE, the resistor value shall be defined by the physical properties of the transistor between BE where the current supply can be defined by the (transconductance \*  $v_{be}$ ) since transconductance is the output current /  $v_{be}$ , cancelling out  $v_{be}$  and giving  $i_c$ .

Base emitter  $R_\pi$

$$R_\pi = \frac{v_{be}}{i_b}, \beta = \frac{i_c}{i_b}, gm = \frac{i_c}{v_{be}}$$

$$gm \frac{1}{\beta} = \left( \frac{i_c}{v_{be}} \right) \left( \frac{i_b}{i_c} \right) = \frac{i_b}{v_{be}} = R_\pi$$

Solving for the transfer function

### KCL (in & out)

$v_{be}$  = small signal across (base emitter)  
 $v_e$  = small signal  $v$  at emitter (to AC GND)  
 $v_c$  = small signal  $v$  out collector (to AC GND)  
 $A_v$  = small signal gain

$$v_{be} = v_{piezo}$$

$$i_c = gm v_{be} = gm v_{piezo}$$

$$v_{out} = -i_c R_c$$

$$A_v = \frac{v_{out}}{v_{in}} = -\frac{gm R_c v_{piezo}}{v_{piezo}} = -gm R_c$$

